Stormwater Regulation Summary: Toronto, ON

Municipal Stormwater Management Technical Profile | Sempergreen USA

1. Regulatory Overview

- Primary driver: Lake Ontario water quality protection and urban flood mitigation.
- Toronto Green Standard (TGS v4) — mandatory Tier 1 for all new development in site plan approval.
- Voluntary Tiers 2-4 offer development charge refunds as incentive for exceeding minimum requirements.
- Toronto was the first major North American city to mandate green roofs (2010 Green Roof Bylaw).
- Ontario Water Resources Act and Clean Water Act provide provincial regulatory framework.

2. Typical Soil Conditions

- Toronto sits on glacial Lake Iroquois deposits — predominantly clay and silt (Halton Till and glaciolacustrine deposits).
- Typical soils: Heavy clay (HSG D) across most of the city; some sandy deposits in the Scarborough Bluffs area.
- Hydrologic Soil Group: Predominantly D — very low infiltration capacity.
- High groundwater table in many waterfront and ravine-adjacent areas further limits infiltration.
- Implication: Infiltration is generally not feasible — Toronto regulations emphasize retention (evapotranspiration via green roofs) and controlled release.

3. Green Roof Requirements

- Green roofs are MANDATORY under the Toronto Green Roof Bylaw (2010, updated 2012).
- Requirement: 20-60% of available roof space must be green roof, scaled by building gross floor area.
- GFA ≤2,000 m²: 20% green roof; GFA 2,000-5,000 m²: gradually increases; GFA >5,000 m²: 60%.
- TGS Tier 1 (mandatory): Retain 5mm rainfall from all roof areas via green roof.
- TGS Tier 2+: 10-25mm retention — incentivized via development charge refunds (can be worth $200K+).
- Eco-Roof Incentive Program: $100/m² rebate (up to $100,000) for green roof installation.

4. TSS Requirements

- Ontario MOECP (Ministry of Environment) requires 80% TSS removal for post-construction stormwater.
- Toronto's Wet Weather Flow Management Policy establishes pollutant reduction targets for combined and storm sewers.
- Green roofs receive TSS removal credit — capturing the design storm volume addresses the highest-TSS fraction of runoff.
- Enhanced treatment (80% TSS, 50% TP) required for all new development under TGS.
- *Specific green roof TSS credit rates should be verified in the current Toronto Stormwater Management Criteria document.*

5. Nature-Based Retention Requirements

- TGS Tier 1 (mandatory): Retain 5mm from all roof areas; on-site retention of 5mm from non-roof areas.
- TGS Tier 2: Retain 10mm from all impervious surfaces.
- TGS Tier 3-4: Retain up to 25mm for maximum development charge credit.
- Toronto uses the Rational Method and Modified Rational Method for peak flow; NRCS methods also accepted.
- CN method applicable for volume-based design; typical urban CN: 90-98 (developed), target 65-75 with green roof.
- Metric units standard: retention volumes specified in mm of rainfall depth over contributing area.

6. Allowable Outflow Rates

- 2-year storm: Post-development peak ≤ pre-development (or 0 increase — depending on local requirements).
- 100-year storm: Safe conveyance with no adverse downstream impacts.
- Toronto Wet Weather Flow Policy targets 5mm/24-hr capture for water quality — effectively limits small-storm discharge to near-zero.
- For reference, typical pre-development rates in Greater Toronto: 20-40 l/s/ha for the 2-year storm.
- *Toronto Region Conservation Authority (TRCA) may impose additional flow restrictions for sites near ravines or watercourses.*
- *Allowable rates are project-specific — confirm with Toronto Water and/or TRCA.*